PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING, SESSION 2026 - 2027

"Design and Implementation of a Cloud-based Robotic Arm Teleoperation System using FPGA-SoC"

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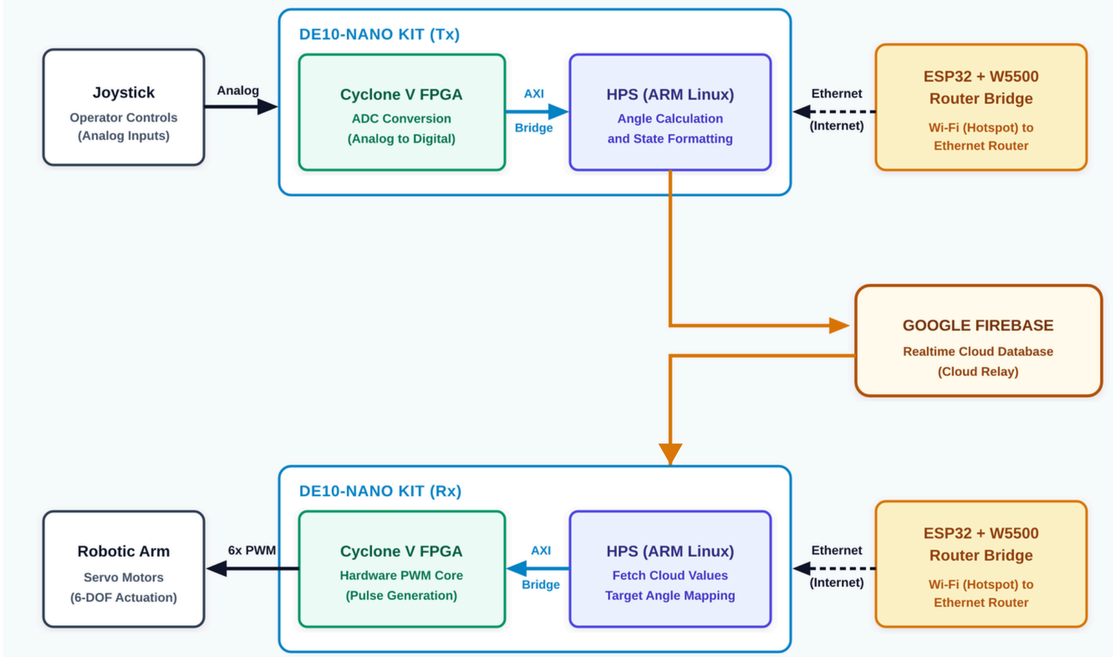
ABSTRACT

Operating robotic machinery in hazardous environments demands real-time, jitter-free control beyond local network cables. This work develops a cloud-based teleoperation framework for a 6-DOF robotic arm using dual Intel DE10-Nano Cyclone V FPGA-SoCs. Operator inputs stream state vectors to Google Firebase via an IoT Ethernet bridge. At the receiver, an ARM Linux processor maps coordinates across a lightweight AXI bridge to dedicated FPGA hardware PWM generators, achieving deterministic, sub-microsecond precision over long distances.

INTRODUCTION

This project presents a cloud-based robotic arm teleoperation system using FPGA-SoC technology. It enables remote control of a robotic arm through cloud communication using Firebase Realtime Database. The DE10-Nano SoC processes received commands through its ARM-based HPS, while the FPGA generates PWM signals to control the servo motors. By offloading deterministic PWM generation directly to dedicated FPGA fabric, the system bypasses Linux scheduling delays, guaranteeing sub-microsecond timing accuracy and jitter-free multi-joint articulation. This integration provides a robust, low-latency platform for remote manipulation in industrial automation and precision manufacturing, such as solar-cell handling.

BLOCK DIAGRAM



APPLICATIONS

- Industrial Automation – Remote pick-and-place operations.
- Remote Inspection – Cloud-based positioning of cameras and sensors.
- Hazardous Environments – Remote robotic manipulation in unsafe areas.

REFERENCES

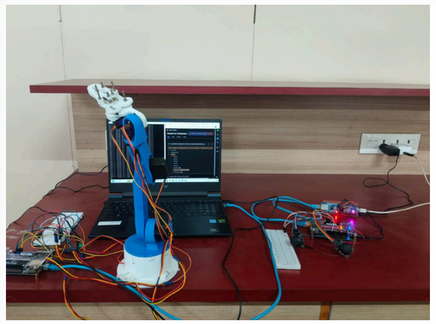
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PROJECT OUTCOME

1. Cable-Free Control: Internet-based robotic control via Firebase.
2. Consistent PWM Timing: FPGA-based PWM generation for reliable servo control.
3. 6-DOF Coordination: Coordinated movement of all six servo motors.





CONCLUSION

This work demonstrates a cloud-based robotic teleoperation system integrating FPGA-SoC control with Firebase communication. By separating cloud-based command handling from FPGA-based PWM generation, the system enables remote operation and coordinated control of a six-DOF robotic arm. Future work includes integrating bilateral haptic feedback and edge-AI-based vision.